

Module/Pathway/Taxon Review: Bacterial Assimilatory Nitrate Uptake and Reduction in *Pseudomonas putida* KT2440

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Target bucket:** kegg:ppu00910 "Nitrogen metabolism" (module area: energy/respiration/inorganic metabolism) **Generic module:** Bacterial assimilatory nitrate uptake and reduction (nitrate → nitrite → ammonium, stopping at ammonium)

1. Executive Summary

The bacterial assimilatory nitrate module is **satisfiable in all three steps** in *P. putida* KT2440. Nitrate import is provided by the nitrate/nitrite porter **NasA (PP_2092, K02575)**, nitrate-to-nitrite reduction is carried out by a fused molybdoenzyme/diflavin assimilatory nitrate reductase **PP_1703 (Q88M71, NasB)**, and the six-electron reduction of nitrite to ammonium is performed by the **NirBD complex (nirB/PP_1705 + nirD/PP_1706)**. The pathway is genomically organized into two loci: a `nasA-nasT-nasS` transporter/regulator cluster (PP_2092/PP_2093/PP_2094) and a `PP_1703-nirB-nirD` reductase cluster, consistent with the classic observation that nitrate-assimilation genes in this lineage are not fully clustered.

The single most important curation finding is that **the actual assimilatory nitrate reductase, PP_1703, is missing from the 20-gene candidate list and is triple-misannotated**. KEGG assigns it K00380 (sulfite reductase flavoprotein), a UniProt ARBA rule over-propagates a periplasmic NapAB (dissimilatory) function, and residual keywords hint at periplasmic localization. All three annotations are refuted here by domain architecture (Molybdopterine guanine dinucleotide + FAD/FMN diaphorase, not a CysJ flavoprotein), by the absence of any Sec/Tat signal peptide (cytoplasmic, not periplasmic), and by genomic context (immediately upstream of nirBD, not part of a nap/nar operon). Because of this misannotation, **automated KEGG module scoring falsely reports the assimilatory module as a gap in ppu**: KEGG's module M00531 requires KOs that are all absent, and KEGG additionally files NirBD under the *dissimilatory* DNRA module M00530 rather than assimilation. The blocker is annotation and database definition, **not biology**.

The remaining 16 of 20 candidate genes are **out of scope** — they belong to GS-GOGAT ammonia assimilation (eight glutamine synthetases, glutamate synthase, glutamate dehydrogenases) and other nitrogen metabolism (carbonic anhydrases, carbamate kinase, nitronate monooxygenases), which the module definition explicitly excludes ("the module stops at ammonium"). KT2440 encodes **no true dissimilatory or denitrification machinery** (no narGHI, napAB, nirK/nirS, nrfA), so the assimilatory module boundary is clean and there is no risk of confusing assimilatory and respiratory routes in this organism. The headline curation actions are: (1) **add and promote PP_1703** to full gene review and correct its annotation; (2) mark all three module steps **covered**; (3) **prune the 16 out-of-scope candidates** from this bucket; and (4) **override automated KEGG scoring** for this taxon.

2. Target-Organism Pathway Definition

Included process. The module covers cytoplasmic assimilatory nitrate metabolism for biosynthesis: (i) uptake of extracellular nitrate (and nitrite) across the inner membrane by a nitrate/nitrite porter; (ii) reduction of nitrate to nitrite by an NAD(P)H-dependent, molybdenum-cofactor assimilatory nitrate reductase; and (iii) the six-electron reduction of nitrite to ammonium by a siroheme/[4Fe-4S] NADH-dependent nitrite reductase (NirBD). The module terminates at ammonium.

Neighboring pathways to keep separate. - **GS-GOGAT ammonia assimilation** (glutamine synthetase, glutamate synthase, glutamate dehydrogenase). This is a *separate* module. Eight of the candidate genes are glutamine synthetases and four more are glutamate synthase/dehydrogenase — all belong here, not in the nitrate module. - **Respiratory nitrate reduction / denitrification** (NarGHI, NapAB, NirK/NirS, Nor, Nos). Absent in KT2440; must never be mapped into the assimilatory module. - **Dissimilatory nitrate reduction to ammonium (DNRA)** (periplasmic NapAB + NrfA). KT2440 lacks NrfA. Note: KEGG idiosyncratically files the *assimilatory* NirBD under its DNRA module M00530, which is a database-definition divergence, not biology. - **Sulfur assimilation / sulfite reduction** (CysJI, CysH). KT2440 encodes a dedicated system (cysI/PP_2371, PP_0860, cysH/PP_2328); the sulfite-reductase KEGG annotation on PP_1703 is a spurious KO assignment, not a real CysJ.

Alternate names / database-specific definitions. - KEGG map: **ppu00910 "Nitrogen metabolism"** (broad overview bucket). - KEGG modules: **M00531** "Assimilatory nitrate reduction, nitrate → ammonia"; **M00530** "Dissimilatory nitrate reduction, nitrate → ammonia." -

The assimilatory nitrate reductase in *Pseudomonas* is historically called **NasB** (in *P. aeruginosa* and *P. putida* genetics), whereas the porter is **NasA** and the sensor/antiterminator regulators are **NasS/NasT**.

3. Expected Step Model

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        nitrate (extracellular)
        |
[Step 1: nitrate import]
NasA / PP_2092 (K02575, NNP/MFS nitrate-nitrite porter)
regulated by NasT/PP_2093 + NasS/PP_2094
        |
        nitrate (cytoplasmic)
        |
[Step 2: nitrate -> nitrite]
PP_1703 / Q88M71 (NasB)
fused Mo-bisMGD + [4Fe-4S] + FAD/FMN NAD(P)H diaphorase
(MISSING from candidate list; mis-annotated K00380)
        |
        nitrite
        |
[Step 3: nitrite -> ammonium]
NirBD: nirB/PP_1705 (K00362, siroheme+[4Fe-4S], large subunit)
      + nirD/PP_1706 (K00363, small subunit)
        |
        ammonium ---> (handoff to GS-GOGAT module; out of scope)

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Step	Expected activity	KT2440 gene(s)	KO	Status
1. Nitrate import	Nitrate/nitrite transmembrane transporter	PP_2092 (nasA)	K02575	Covered
2. Nitrate → nitrite	Assimilatory NAD(P)H nitrate reductase (fused Mo/diflavin)	PP_1703 (nasB)	mis-KO'd K00380 → should be assimilatory NR	Covered (annotation gap)
3. Nitrite → ammonium	Assimilatory NADH nitrite reductase (siroheme)	PP_1705 (nirB) + PP_1706 (nirD)	K00362 + K00363	Covered

The generic module offers alternative architectures for step 2 (fused diflavin NasB vs. split NasBC vs. ferredoxin NarB). KT2440 uses the **fused diflavin NasB** implementation. For step 1 it offers NNP porter vs. NrtABCD ABC importer; KT2440 uses the **NNP porter** (no NrtABCD KOs present).

4. Candidate Genes and Evidence

4.1 In-scope, high-confidence genes

PP_2092 — nasA — Nitrate/nitrite transporter (K02575). UniProt Q88L43, 411 aa, MFS/NNP-family porter with UniProt keyword "Nitrate assimilation," predicted inner-membrane transmembrane transporter. It is the **sole nitrate porter** in the genome — no second NNP/NarK porter (K02598) and no NrtABCD ABC importer (K15576–K15579) exist. It is flanked by NasT (PP_2093, K07183, ANTAR-domain response-regulator/antiterminator) and NasS (PP_2094, K22067, nitrate/nitrite-binding sensor), forming an intact nitrate-sensing regulon. *Curation note:* clean, high-confidence. Evidence type: sequence/family + genomic context + regulon synteny.

PP_1703 — nasB — Assimilatory nitrate reductase (currently mis-KO'd K00380). UniProt Q88M71, 1341 aa. Domain content: Pfam PF00384 (Molybdopterin) + PF04879 (Molybdop_Fe4S4) + PF01568 (Molybdopterin dinucleotide-binding), plus PF00667

(FAD_binding_1) and PF00258 (Flavodoxin_1). Cofactors: FAD, FMN, Mo-bis(molybdopterin guanine dinucleotide), [4Fe-4S]. UniProt places it in the "oxidoreductase family, NasA/NapA/NarB subfamily" (InterPro "CT_Nitrate-R-NapA-like"). It lies in an operon immediately upstream of nirB (PP_1705) and nirD (PP_1706). *This gene is absent from the 20-candidate list and must be added.* Evidence type: domain architecture + genomic context + lineage genetics (KT2442 nasB,

).

P 10852866

Promote to full review.

PP_1705 — nirB — Nitrite reductase [NAD(P)H] large subunit (K00362, EC 1.7.1.15). UniProt Q88M69 with confirmed siroheme and [4Fe-4S] binding-site annotations (residues ~646/652/686/690). Carries the catalytic six-electron reduction of nitrite to ammonium. No competing NO-forming (nirK/nirS) or DNRA (nrfA) nitrite reductase exists, so NirBD is the **sole nitrite-reduction route**. High-confidence. Evidence type: sequence + cofactor sites + absence of alternatives + functional data in *P. putida* Y-9.

PP_1706 — nirD — Nitrite reductase small subunit (K00363). UniProt Q88M68. Partner subunit of NirB. High-confidence.

4.2 Out-of-scope candidates (belong to GS-GOGAT or other N-metabolism)

Gene	Locus	KO	Assignment	Correct home
glnA	PP_5046	K01915	Glutamine synthetase	Ammonia assimilation
PP_3148	PP_3148	K01915	Glutamine synthetase	Ammonia assimilation
PP_4399	PP_4399	K01915	Glutamine synthetase	Ammonia assimilation
PP_4547	PP_4547	K01915	Glutamine synthetase	Ammonia assimilation
puuA-I	PP_2178	K01915	Glutamate-putrescine ligase	Polyamine/GS family
puuA-II	PP_5299	K01915	Glutamate-putrescine ligase	Polyamine/GS family
spuB	PP_5183	K01915	Glutamylpolyamine synthetase	Polyamine metabolism
spuI	PP_5184	K01915	Glutamylpolyamine synthetase	Polyamine metabolism
gltB	PP_5076	K00265	Glutamate synthase large chain	GS-GOGAT
gltD	PP_5075	K00266	Glutamate synthase small chain	GS-GOGAT
gdhA	PP_0675	K00262	Glutamate dehydrogenase	GS-GOGAT-adjacent
gdhB	PP_2080	K15371	NAD-glutamate dehydrogenase	Glutamate catabolism (ppu00430)
cynT	PP_0100	K01673	Carbonic anhydrase	Other
PP_3392	PP_3392	K28956	γ -carbonic anhydrase	Other
arcC	PP_0999	K00926	Carbamate kinase	Arginine/urea
yrpB	PP_3827	K00459	Nitronate monooxygenase	Nitroalkane detox
PP_0430	PP_0430	K00459	Nitronate monooxygenase (uncharacterized)	Nitroalkane detox

These loci are correctly part of KEGG ppu00910 "Nitrogen metabolism" as a broad overview map, but **do not belong to the assimilatory nitrate module**, whose scope explicitly ends at ammonium. [gltB](#) (PP_5076) is notable: although out-of-scope for the module, it is functionally

coupled — inactivation of *gltB* abolishes *nasB* expression in KT2442 (P 10852866),
evidencing a regulatory link between GOGAT and nitrate assimilation.

5. Gaps, Ambiguities, and Likely Over-Annotations

5.1 The missing reductase (the central gap). Step 2 has no representative in the candidate list. The true enzyme, PP_1703, is present in the genome and functional in the lineage but was omitted because it is not KO-mapped to any nitrate-reductase KO. This is a **metadata gap, not a biological gap**. Whole-genome scan: the canonical assimilatory nitrate reductase KOs K00372/K00360 are **absent**; K00367 (ferredoxin-nitrate reductase, *narB*) is **absent**. PP_1703 is the fused *NasB* implementation and must be recognized manually.

5.2 Triple mis-annotation of PP_1703 — all refuted. - **KEGG K00380 (sulfite reductase flavoprotein alpha, CysJ).** Refuted: PP_1703 carries a molybdenum cofactor, which CysJ never has; and KT2440 has an independent, complete sulfite-reduction system (*cysI*/PP_2371 siroheme hemoprotein, PP_0860 flavoprotein, *cysH*/PP_2328 APS reductase), so PP_1703 is not required for sulfur assimilation. - **UniProt ARBA "periplasmic assimilatory/dissimilatory NapAB."** Refuted: PP_1703 has **no Sec or Tat signal peptide** (no FT SIGNAL, no FT TRANSIT, no subcellular-location feature). Periplasmic NapA is Tat-exported and ~750–830 aa with no fused flavin domain; PP_1703 is 1341 aa with a C-terminal FAD/flavodoxin NAD(P)H diaphorase (residues ~819–957 and ~981–1191) and an N-terminal molybdoenzyme [4Fe-4S] motif (CPYCGVGCG). It is a **cytoplasmic fused NasB**, not periplasmic NapA. Confirming this, *napA/napB* (K02567/K02568) and *narGHI* are entirely absent from the *ppu* genome. - **Residual periplasmic/localization keywords.** Refuted by the same absence of any export signal.

5.3 KEGG module-definition divergence. KEGG module M00531 DEFINITION = (K00367,K10534,(K00372-K00360)) (K00366,K17877,(K26139+K26138),K00361). **None** of these KOs is assigned to any *ppu* gene, so KEGG lists M00531 as *not present* in *ppu*. Two problems: (i) PP_1703 is mis-KO'd away from any of these; (ii) the nitrite branch of M00531 demands a ferredoxin/NirA-type reductase (K00366 etc.), whereas KT2440's real nitrite reductase is **NirBD (K00362+K00363)**, which KEGG places in the *dissimilatory* module M00530 (DNRA). Automated module scoring therefore **falsely flags a gap**. Manual override is required.

5.4 Decoy transporter. PP_0208 (Q88RC4) is annotated "Nitrate ABC transporter permease" but sits in operon PP_0207 (K15553) – PP_0208 (K15554) – PP_0209 tauB-I (K15555): this is the **ssuABC/tauABC aliphatic-sulfonate ABC transporter** family (with a nearby sulfonate-binding protein and ferredoxin), **not** a nitrate importer. Do not count it toward step 1.

5.5 Paralog / broad-mapping ambiguities. Eight loci collapse onto K01915 (glutamine synthetase); this broad GS mapping is a classic source of over-propagation into "nitrogen metabolism" buckets. Two loci (yrbP, PP_0430) share K00459 (nitronate monooxygenase). None affects the nitrate module, but curators should not let the sheer count of GS/other-N genes obscure the fact that only 3 (now 4 with PP_1703) candidates are truly in-scope.

6. Module and GO-Curation Recommendations

Module step	Recommended status	Basis
1. Nitrate import (NNP porter)	covered	PP_2092 nasA (K02575), sole porter, NasST-regulated
2. Nitrate → nitrite (fused NasB)	covered — requires metadata fix	PP_1703 present & functional; add to bucket, correct KO/annotation
3. Nitrite → ammonium (NirBD)	covered	PP_1705 nirB (K00362) + PP_1706 nirD (K00363)
Alt: NrtABCD ABC importer	not_expected_in_target_taxon	No K15576–K15579 in genome
Alt: ferredoxin NarB reductase	not_expected_in_target_taxon	No K00367
Respiratory/ denitrification neighbors	not_expected_in_target_taxon	No narGHI, napAB, nirK/nirS, nrfA

Module-level actions. 1. Add PP_1703 (Q88M71) to the kegg:ppu00910 assimilatory-nitrate **bucket** as the step-2 enzyme (NasB). This is the single most important edit. 2. Mark the whole module **satisfiable / covered (3/3)** for KT2440, with an explicit note that automated KEGG M00531 scoring must be **overridden** because (a) PP_1703 is mis-KO'd to K00380 and (b) KEGG maps the organism's actual nitrite reductase NirBD into the dissimilatory M00530. 3. **Prune the**

16 out-of-scope candidates from the assimilatory-nitrate module (retain them under the broader ppu00910 overview / ammonia-assimilation module as appropriate). 4. Flag **PP_0208** as a mislabeled sulfonate transporter (annotation-correction candidate), not a nitrate importer. 5. Generic module boundaries are **correct** for this organism (the module rightly stops at ammonium and excludes GS-GOGAT). No boundary revision is needed; the issue is entirely KT2440 metadata mapping — status **module_needs_revision** applies only to the *local KO/candidate metadata*, not to the generic module document.

GO-curation notes. PP_1703 should carry GO:0008940 (nitrate reductase activity) / GO:0042126 (nitrate metabolic process) / GO:0042128 (nitrate assimilation) and a **cytoplasm** localization (GO:0005737) rather than periplasm. Its erroneous sulfite-reductase and periplasmic-NapAB associations should be removed. No new GO term requests appear necessary — existing terms for assimilatory nitrate reductase, NADH nitrite reductase, and nitrate transmembrane transporter cover all three steps.

7. Genes to Promote to Full Review

Gene	Locus	UniProt	Why promote
PP_1703 (nasB)	PP_1703	Q88M71	Highest priority. Missing from candidate list; triple mis-annotated; is the true step-2 assimilatory nitrate reductase. Full fetch-gene review to correct KO (away from K00380), remove periplasmic NapAB propagation, set cytoplasmic localization, and register as NasB.
PP_2092 (nasA)	PP_2092	Q88L43	Confirm as sole nitrate porter; document NasST regulon linkage (PP_2093/PP_2094).
PP_1705 (nirB)	PP_1705	Q88M69	Confirm siroheme/[4Fe-4S] assimilatory nitrite reductase; ensure it is not mis-filed as dissimilatory DNRA.
PP_1706 (nirD)	PP_1706	Q88M68	Partner subunit; confirm assimilatory role.
PP_0208	PP_0208	Q88RC4	Annotation correction: reassign from "nitrate ABC transporter" to ssuABC/tauABC sulfonate transporter.

8. Evidence and Open Questions

Direct experimental support (target lineage). The strongest species-level evidence is genetic: a transposon insertion mapping to the **assimilatory nitrate reductase gene *nasB*** in the near-isogenic strain **KT2442** (PMID: 10852866). That study's own words — "*is demonstrated to bear the transposon within the assimilatory nitrate reductase gene (*nasB*) of *P. putida* KT2442*" and "*Genetic evidence as well as sequence analyses of the DNA regions flanking *nasB* suggest that the genes required for nitrate assimilation are not clustered*" — provide direct proof that the KT2440 lineage encodes a functional assimilatory nitrate reductase and explain why the porter (PP_2092) and reductase (PP_1703) loci are genomically separate. KT2442 is essentially isogenic to KT2440 (a restriction-modification variant), so transfer of this conclusion to KT2440 is **strong**.

Cross-strain / cross-species support (moderate transfer). NirBD's assimilatory nitrite-to-ammonium role is functionally demonstrated in *P. putida* Y-9, where "*The nirBD that encodes nitrite reductase had an important role in strain growth and ammonium production*" (PMID: 32506044; see also PMID: 41980646). Because Y-9 is a different *P. putida* strain that also runs DNRA/denitrification, transfer of the *assimilatory* interpretation to KT2440 is **moderate**; however, KT2440 lacks the dissimilatory machinery, so in KT2440 NirBD can only be assimilatory. The genetic distinctness of assimilatory (*nas*) and dissimilatory (*nar*) nitrate reductases in the genus is established in *P. aeruginosa* (PMID: 6775047) — **weak-to-moderate** transfer, useful for the general principle that Nas ≠ Nar and that both require molybdenum.

Inferred from homology / database evidence. The domain architecture of PP_1703 (Mo-bisMGD + [4Fe-4S] + FAD/FMN diaphorase), the absence of a signal peptide, the KO assignments of PP_2092/PP_1705/PP_1706, the NasST regulon, and the absence of all dissimilatory/denitrification and ABC-importer KOs derive from UniProt/KEGG/Pfam/InterPro annotations and whole-genome KO scans. These are **homology/sequence inferences**, strong for family assignment but not a substitute for direct enzymology in KT2440 itself.

Open questions / resolving experiments. 1. **Direct enzymology on PP_1703:** heterologous expression and NAD(P)H:nitrate reductase assay to confirm activity and cofactor complement; would definitively retire the sulfite-reductase and NapAB misannotations. 2. **Subcellular fractionation** of PP_1703 to confirm cytoplasmic localization (the signal-peptide analysis predicts cytoplasm). 3. **Growth phenotyping of KT2440 (not KT2442)** on nitrate as sole N source, with ΔPP_1703, ΔPP_2092, and ΔnirBD knockouts, to confirm each step's necessity in the exact target strain. 4. **RT-PCR / RNA-seq** to verify the PP_1703–nirB–nirD operon and NasST-

dependent, nitrate-inducible expression. 5. **Expert/database question:** should KEGG's placement of NirBD in the dissimilatory module M00530 be flagged for taxa (like KT2440) that lack DNRA machinery, so automated scoring stops mis-assigning it?

9. Key References

PMID	Title (abbrev.)	Relevance
10852866	<i>Inactivation of gltB abolishes expression of the assimilatory nitrate reductase gene (nasB) in P. putida KT2442</i>	Direct genetic evidence for assimilatory nitrate reductase nasB in the KT2440 lineage; shows nas genes are not fully clustered and links GOGAT to nasB expression.
32506044	<i>Nitrate assimilation, DNRA, and denitrification coexist in P. putida Y-9 under aerobic conditions</i>	Functional evidence that NirBD reduces nitrite to ammonium and supports growth/ammonium production.
41980646	<i>The nirB and nirD genes are essential for reduction of nitrite in P. putida Y-9 under aerobic conditions</i>	Confirms NirBD requirement for nitrite reduction (assimilatory and dissimilatory) under aerobic conditions.
6775047	<i>The assimilatory and dissimilatory nitrate reductases of P. aeruginosa are encoded by different genes</i>	Establishes that Nas (assimilatory) and Nar (dissimilatory) reductases are genetically separate in <i>Pseudomonas</i> ; both require molybdenum.

Summary verdict

Assimilatory nitrate module = SATISFIABLE (3/3) in *P. putida* KT2440. Import = PP_2092 (nasA), reduction = PP_1703 (nasB, must be added and corrected), nitrite reduction = PP_1705/PP_1706 (nirBD). The only obstacles are (1) PP_1703's omission and triple misannotation and (2) KEGG's module definitions — both curation/metadata problems, not biological gaps. No dissimilatory or denitrification machinery exists, so the module boundary is clean and requires no revision.

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